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I: Data Acquisition

1. The user places their finger on the MAX30102 sensor.
2. The sensor will capture the user's change in blood volume in their veins with IR LED, and it will capture changes in haemoglobin in blood with Red LED (majorly).
3. This data is captured by ESP-8266, and further processed to get an estimated heart rate and peripheral capillary oxygen saturation (SpO2) value.

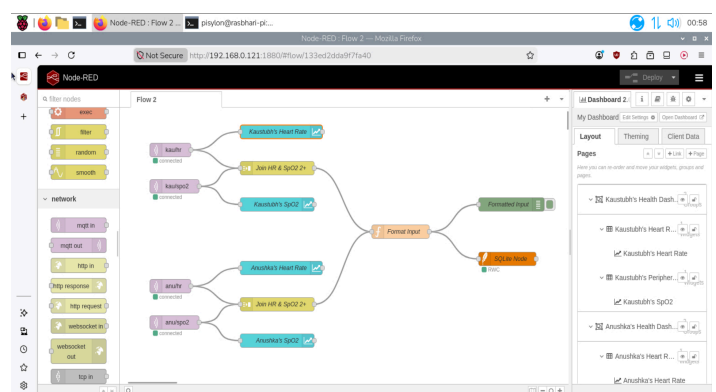
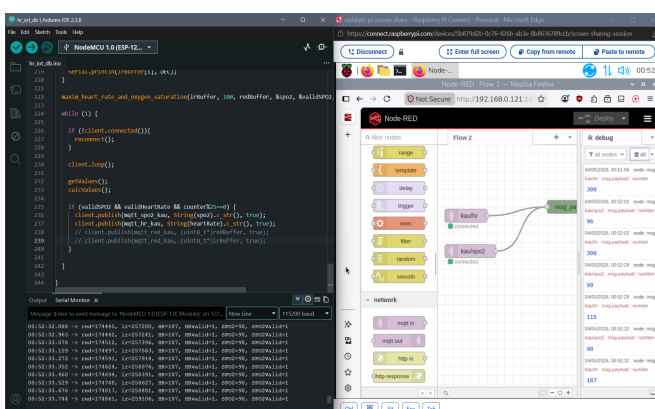
II: Raspberry-Pi Server

1. A Raspberry-Pi 2 Model B is used as a server for storing the data captured by ESP-8266 (although it is not straight away stored).
2. Connection to RPi was made majorly with the help of Raspberry-Pi Connect.
3. To send the data to this server, MQTT is used with the help of a package called "Mosquitto". It runs in the background and listens to port 1883 for any messages.

III: Node-RED Visualization

Node-RED is a flow-based, low-code development tool designed for visual programming, primarily aimed at wiring together hardware devices, APIs, and online services, especially in the context of the Internet of Things (IoT).

1. It subscribes to topics which are publishing, and with the help of Mosquitto running in the background, it captures the messages.
2. For the dashboard, a package called 'node-red-dashboard' was used.
3. Data from Node-RED was sent to the local RPi server (on top of which Node-RED was running) and the data was logged using SQLite.



IV: SQLite (sqlite3)

The package which was used is called 'sqlite3'; it can be installed from the 'apt' repository.

1. A 'sqlite3' database file was created and a schematic was also created inside of it.
2. As input kept coming from ESP-8266, Node-RED sent it to the local storage, forming a database. Node-RED had to be configured to do so.

```
pisylon@rasbhari-pi: /var/www/html/database
File Edit Tabs Help

pisylon@rasbhari-pi:~$ cd /var/www/html/database/
pisylon@rasbhari-pi:/var/www/html/database$ ls
max30102.db  phpliteadmin.config.sample.php  phpliteadmin.php  readme.md
pisylon@rasbhari-pi:/var/www/html/database$ sqlite3 max30102.db
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> DROP TABLE heart_rate_data
...;
sqlite>
sqlite> CREATE TABLE heart_rate_data (
(x1...> id INTEGER PRIMARY KEY AUTOINCREMENT,
(x1...> person TEXT,
(x1...> bpm REAL,
(x1...> spo2 REAL,
(x1...> timestamp DATETIME DEFAULT CURRENT_TIMESTAMP
(x1...> );
sqlite>
sqlite>
sqlite>
```

```
pisylon@rasbhari-pi: /var/www/html/database
File Edit Tabs Help

13|Kaustubh|166.0|100.0|2026-05-04 16:11:46
12|Kaustubh|166.0|100.0|2026-05-04 16:11:46
11|Kaustubh|166.0|100.0|2026-05-04 16:11:43
10|Kaustubh|166.0|100.0|2026-05-04 16:11:43
9|Kaustubh|166.0|100.0|2026-05-04 16:11:41
sqlite> SELECT * FROM heart_rate_data ORDER BY id DESC LIMIT 5;
13|Kaustubh|166.0|100.0|2026-05-04 16:11:46
12|Kaustubh|166.0|100.0|2026-05-04 16:11:46
11|Kaustubh|166.0|100.0|2026-05-04 16:11:43
10|Kaustubh|166.0|100.0|2026-05-04 16:11:43
9|Kaustubh|166.0|100.0|2026-05-04 16:11:41
sqlite> SELECT * FROM heart_rate_data ORDER BY id DESC LIMIT 5;
25|Kaustubh|125.0|99.0|2026-05-04 16:12:48
24|Kaustubh|150.0|99.0|2026-05-04 16:12:48
23|Kaustubh|150.0|100.0|2026-05-04 16:12:45
22|Kaustubh|136.0|100.0|2026-05-04 16:12:45
21|Kaustubh|136.0|94.0|2026-05-04 16:12:40
sqlite> SELECT * FROM heart_rate_data ORDER BY id DESC LIMIT 5;
31|Kaustubh|187.0|100.0|2026-05-04 16:12:55
30|Kaustubh|150.0|100.0|2026-05-04 16:12:55
29|Kaustubh|150.0|95.0|2026-05-04 16:12:53
28|Kaustubh|107.0|95.0|2026-05-04 16:12:53
27|Kaustubh|107.0|98.0|2026-05-04 16:12:50
sqlite>
```

V: Bash Script

1. A bash script was written so that the users could easily interact with the database, and gain some insights from the captured sensor readings.
2. The bash script consists of a menu and a few options. The user has to select the options, and the SQL query will be run for them (without having to type it themselves) by the bash script.

```
pisylon@rasbhari-pi:/var/www/html/database$ ./queries_script.sh
---- WELCOME ----
Please select any query by typing its sr. no.

1. View heart rate data (last 50 entries) of...
2. Calculate average heart rate of...
3. Compare average heart of users.
4. View oxygen saturation (SpO2) data (last 50 entries) of...
5. Calculate average SpO2 of...
6. Compare average SpO2 of all users.
7. Get Minimum, Maximum, and Average of...

Enter sr. no. of query: 2

Calculate average HR for: Kaustubh

AVG(bpm)
-----
143.6363636364

Do you want to continue? (Y/n): n
Exiting program...
pisylon@rasbhari-pi:/var/www/html/database$
```

```
pisylon@rasbhari-pi:/var/www/html/database$ ./queries_script.sh
---- WELCOME ----
Please select any query by typing its sr. no.

1. View heart rate data (last 50 entries) of...
2. Calculate average heart rate of...
3. Compare average heart of users.
4. View oxygen saturation (SpO2) data (last 50 entries) of...
5. Calculate average SpO2 of...
6. Compare average SpO2 of all users.
7. Get Minimum, Maximum, and Average of...

Enter sr. no. of query: 
```

VI: Populated Dashboard

1. The RPi and ESP-8266 communicate very fast, and the values are updated almost instantly on the Node-RED dashboard and RPi terminal.
2. The following chart shows the populated dashboard after taking reading for 1 hour

